Customer Churn Prediction System

Complete Project Report — End-to-End ML Pipeline

7,021	24	0.84	200	0.90
Customers Analysed	Features Used	ROC-AUC Score	Decision Trees	CV AUC Score

What This Report Covers

Dataset & Features | Data Preprocessing | Feature Engineering | SMOTE Balancing
Exploratory Data Analysis (9 Charts) | SQL Business Analytics | ML Model Training
Random Forest Deep-Dive | Model Evaluation | Feature Importance | Business Impact

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CustomerPulse AI — Data Science Portfolio Project

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1. Executive Summary

This report documents the **CustomerPulse AI — Customer Churn Prediction System**, a complete, production-grade machine learning pipeline built to identify telecom customers who are likely to cancel their subscription. The system was trained on the IBM Telco Customer Churn dataset containing **7,021 customers** and **20 original features**, augmented with **5 engineered features** to reach 24 total predictors.

Metric	Value	Metric	Value
Total Customers	7,021	Best Model	Random Forest
Churn Rate	26.45%	ROC-AUC	0.8362
Features Used	24 (19+5 eng)	5-Fold CV AUC	0.9012 ± 0.0245
Training Set	5,617 (80%)	Recall (Churners)	70.7%
Test Set	1,404 (20%)	F1-Score (Churn)	0.6165
SMOTE Applied	Yes (balanced)	Monthly Revenue Risk	$138,529
Pipeline Runtime	8.6 seconds	Trees in Forest	200
EDA Figures Generated	9 charts	Accuracy	77.4%

Key Findings: The Random Forest model achieved a ROC-AUC of 0.8362 and a 5-fold cross-validation AUC of 0.9012 — correctly identifying 70.7% of all customers who will actually churn. The most critical finding: **month-to-month customers on fiber optic plans with electronic check payment churns at 54.48%** — the single highest-risk customer segment. Targeting the top 630 highest-risk customers with a $50 retention incentive each is estimated to recover $70,200 in annual revenue against a campaign cost of $31,500.

2. Project Architecture & File Structure

The system follows a modular, sequential pipeline architecture. Each stage is encapsulated in its own Python module inside the src/ directory, with main.py orchestrating execution from start to finish.

```
Customer_Churn_Prediction_System/
main.py ← Pipeline orchestrator (runs all 5 stages)
dashboard/
app.py ← Streamlit dashboard (5 interactive pages)
src/
preprocessing.py ← Stage 1: data cleaning, encoding, scaling
eda.py ← Stage 3: 9 matplotlib/seaborn charts
sql_integration.py ← Stage 4: SQLite database + 7 business queries
models.py ← Stage 5: Logistic Regression, RF, XGBoost
data/
WA_Fn-UseC_-Telco-Customer-Churn.csv ← Raw IBM Telco dataset
models/
best_model.pkl ← Serialised Random Forest (pickle)
model_results.json ← All model metrics saved as JSON
scaler.pkl ← StandardScaler fitted on training data
feature_names.pkl ← Ordered list of 24 feature names
reports/figures/ ← 13 PNG charts (9 EDA + 4 model)
churn_data.db ← SQLite database with all customer records
requirements.txt ← All Python dependencies
```

Stage	Module	What It Does	Output
1 — Preprocessing	preprocessing.py	Loads CSV, cleans nulls, encodes categoricals, scales numerics, applies SMOTE	X_train, X_test arrays
2 — Feat. Engineering	preprocessing.py	Creates RiskScore, ChargesPerTenure, ServiceCount, TenureGroup, HighValueFlag	24 total features
3 — EDA	eda.py	Generates 9 matplotlib/seaborn charts covering all key distributions and relationships	9 PNG files
4 — SQL Analytics	sql_integration.py	Stores data in SQLite, runs 7 business queries using native sqlite3	Printed tables + DB
5 — Model Training	models.py	Trains 3 models, tunes RF, runs 5-fold CV, saves best model as pickle	best_model.pkl + JSON

3. Dataset Overview — 7,021 Customers

The raw data comes from the **IBM Sample Data — Telco Customer Churn** dataset. It represents one quarter of customer activity for a fictional California-based telecom company. Each row is a unique customer; each column describes an aspect of their service, account, or demographic profile.

Property	Value	Notes
Source	IBM Developer / Kaggle	Widely used benchmark dataset for churn prediction
Total Rows	7,043 raw → 7,021 clean	22 rows dropped (TotalCharges = whitespace / null)
Total Columns (raw)	21	customerID (dropped) + 19 features + 1 target
Target Variable	Churn (Yes/No)	Encoded as 1=Churn, 0=No Churn for modelling
Churn = Yes	1,857 (26.45%)	Minority class — imbalanced dataset
Churn = No	5,164 (73.55%)	Majority class
Numeric Features	3 original	tenure, MonthlyCharges, TotalCharges
Categorical Features	16 original	Contract, InternetService, PaymentMethod, etc.
Binary Features	9 original	gender, SeniorCitizen, Partner, Dependents, etc.
After Engineering	24 features total	19 processed originals + 5 engineered
Time Period	Not specified	Cross-sectional snapshot — one point in time

Important: **Class Imbalance Warning:** With only 26.45% positive class (churners), a naive model that always predicts 'No Churn' would achieve 73.55% accuracy — making accuracy a misleading metric. This is why SMOTE oversampling and ROC-AUC are used instead of raw accuracy for model selection.

4. All 24 Features Explained

Complete reference table of every feature used by the model, including type, range, encoding, and relationship to churn.

Feature Name	Type	Range/Encoding	Description	Importance	Risk Tier
RiskScore	Engineered	0.0-1.0	Composite: 0.4×contract+0.3×charges+0.3×tenure risk	#1 (19.3%)	CRITICAL
ChargesPerTenure	Engineered	0.0-38.25	MonthlyCharges÷(tenure+1). High=new+expensive	#2 (14.1%)	CRITICAL
Contract	Categorical	0/1/2	M2M=0, 1yr=1, 2yr=2	#3 (12.3%)	CRITICAL
MonthlyCharges	Numeric	$19.20-$114.74	Raw monthly bill	#4 (8.5%)	HIGH
tenure	Numeric	0-72 months	Months with company	#5 (8.1%)	HIGH
TotalCharges	Numeric	$19.75-$8,040	Cumulative billing	#6 (6.7%)	HIGH
InternetService	Categorical	0/1/2	DSL=0, Fiber=1, None=2	#7 (4.7%)	MEDIUM
OnlineSecurity	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#8 (3.9%)	MEDIUM
PaymentMethod	Categorical	0/1/2/3	AutoBank=0,eCheck=1,Mail=2,Card=3	#9 (3.4%)	MEDIUM
TechSupport	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#10 (3.0%)	MEDIUM
ServiceCount	Engineered	0-8	Sum of all active add-on services	#11	LOW
TenureGroup	Engineered	0-5	6 bins: 0-12,13-24,25-36,37-48,49-60,61-72 months	#12	LOW
HighValueFlag	Engineered	0 or 1	1 if MonthlyCharges>$65 AND tenure>12 months	#13	LOW
OnlineBackup	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#14	LOW

DeviceProtection	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#15	LOW
StreamingTV	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#16	LOW
StreamingMovies	Categorical	0/1/2	No=0, No Internet=1, Yes=2	#17	LOW
PaperlessBilling	Binary	0 or 1	No=0, Yes=1	#18	LOW
Partner	Binary	0 or 1	No=0, Yes=1	#19	LOW
Dependents	Binary	0 or 1	No=0, Yes=1	#20	LOW
SeniorCitizen	Binary	0 or 1	Not Senior=0, Senior=1	#21	LOW
PhoneService	Binary	0 or 1	No=0, Yes=1 (91% have it)	#22	MINIMAL
MultipleLines	Categorical	0/1/2	No=0, No Phone=1, Yes=2	#23	MINIMAL
gender	Binary	0 or 1	Female=0, Male=1	#24 (~0%)	NONE

5. Data Preprocessing Pipeline

The preprocessing pipeline transforms raw CSV data into a clean, encoded, and scaled numerical matrix ready for machine learning. All steps are implemented in src/preprocessing.py using scikit-learn pipelines.

Step	Operation	Detail	Rows/Cols Affected
1	Load CSV	pandas.read_csv() on WA_Fn-UseC_-Telco-Customer-Churn.csv	7,043 rows × 21 cols
2	Drop customerID	Unique identifier — no predictive value, removed immediately	1 column dropped
3	Fix TotalCharges	22 rows had ' ' (whitespace). Converted to float, nulls dropped	22 rows removed
4	Encode Target	Churn: 'Yes'→1, 'No'→0. Result: 1,857 positives (26.45%)	1 column transformed
5	Label Encode	All object-type columns encoded with LabelEncoder to integers	16 columns encoded
6	Feature Engineer	5 new columns created: RiskScore, ChargesPerTenure, ServiceCount, TenureGroup, HighValueFlag	5 columns added
7	Train-Test Split	stratify=Churn, test_size=0.20, random_state=42. Preserves 26.45% ratio in both sets	5,617 train / 1,404 test
8	StandardScaler	Fit on X_train only, transform both X_train and X_test. Prevents data leakage	3 numeric cols scaled
9	SMOTE	Applied to X_train only after split. Generates synthetic minority samples	Training set balanced

Engineering Note: **Data Leakage Prevention:** The StandardScaler is fit ONLY on the training set (X_train), then applied to both X_train and X_test using the same scaling parameters. Similarly, SMOTE is applied ONLY to X_train after the split. This ensures that the test set represents truly unseen data with no information from the training process.

6. Feature Engineering — 5 New Variables

Feature engineering is the process of using domain knowledge to create new, more predictive variables from existing ones. Your project created 5 engineered features that became the most important predictors — outperforming all 19 original columns.

Feature 1: RiskScore (Importance: 19.3% — #1 Overall)

A composite churn risk index combining the three strongest individual predictors into one signal:

```
RiskScore = 0.4 × contract_risk + 0.3 × charges_norm + 0.3 × tenure_risk

contract_risk = 1.0 if Month-to-Month | 0.5 if 1-Year | 0.0 if 2-Year
charges_norm = (MonthlyCharges - 19.20) / (114.74 - 19.20) # normalised 0→1
tenure_risk = 1.0 - (tenure / 72) # 1=new, 0=loyal

Range: 0.00 (safest) → 1.00 (highest risk)
Customer #4865 example: 0.4×1.0 + 0.3×1.0 + 0.3×0.972 = 0.992
```

Feature 2: ChargesPerTenure (Importance: 14.1% — #2 Overall)

Measures how much a customer pays relative to how long they have been with the company:

```
ChargesPerTenure = MonthlyCharges / (tenure + 1)

+1 prevents division-by-zero for brand-new customers (tenure = 0)

Low value (e.g. 1.2): Long-term, value-seeking customer → LOW churn risk
High value (e.g. 38.2): New customer paying maximum rates → HIGH churn risk
```

Feature 3: ServiceCount

```
ServiceCount = PhoneService + MultipleLines + OnlineSecurity + OnlineBackup
+ DeviceProtection + TechSupport + StreamingTV + StreamingMovies

Range: 0 → 8. More services = more invested = harder to switch = lower churn.
```

Feature 4: TenureGroup

```
TenureGroup = pd.cut(tenure, bins=[0,12,24,36,48,60,72], labels=[0,1,2,3,4,5])

Group 0 (0-12 mo): 47.4% churn rate ← CRITICAL retention window
Group 1 (13-24 mo): 28.7% churn rate
```

```
Group 2 (25-48 mo): 20.4% churn rate
Group 3 (49+ mo): 9.5% churn rate
```

Feature 5: HighValueFlag

```
HighValueFlag = 1 if (MonthlyCharges > 65) AND (tenure > 12) else 0

Identifies long-term, high-paying customers — the VIP segment.
These customers are highest-priority for proactive retention campaigns.
```

7. SMOTE — Handling Class Imbalance

The raw training data has 73.55% 'No Churn' and 26.45% 'Churn' — a significant class imbalance. Without correction, the model would be biased towards predicting 'No Churn' for almost every customer. SMOTE (Synthetic Minority Oversampling Technique) fixes this by generating realistic synthetic churn examples.

Stage	Churn = Yes	Churn = No	Ratio	Note
Original dataset (7,021 rows)	1,857 (26.4%)	5,164 (73.6%)	1:2.8	Imbalanced
After train/test split — Train	1,485 (26.4%)	4,132 (73.6%)	1:2.8	Still imbalanced
After SMOTE — Train only	4,132 (50.0%)	4,132 (50.0%)	1:1.0	Balanced for training
Test set (untouched)	372 (26.5%)	1,032 (73.5%)	1:2.8	Real-world distribution

How SMOTE Creates Synthetic Samples:

```
For each minority (Churn=Yes) sample:
1. Find its k=5 nearest neighbours in feature space (also Churn=Yes)
2. Randomly pick one of those 5 neighbours
3. Randomly interpolate between the original and neighbour:
synthetic_point = original + random(0,1) × (neighbour - original)

Example (simplified):
Real customer A: RiskScore=0.85, MonthlyCharges=95.00, tenure=3
Nearest neighbour: RiskScore=0.91, MonthlyCharges=102.00, tenure=5
Random factor = 0.4
Synthetic: RiskScore=0.87, MonthlyCharges=97.80, tenure=3.8

SMOTE generates 2,647 synthetic churners to balance the training set.
The test set is NEVER oversampled — it reflects real-world distribution.
```

SMOTE vs Duplication: **Why not just duplicate existing samples?** Simple duplication (oversampling) causes overfitting — the model memorises specific churners. SMOTE creates *interpolated* samples that explore the feature space between real churners, teaching the model to generalise the characteristics of churn rather than memorise examples.

8. Exploratory Data Analysis — 9 Charts

Before training any model, the pipeline runs a comprehensive Exploratory Data Analysis (EDA) to understand distributions, relationships, and churn patterns in the data. All 9 charts are generated by src/eda.py using matplotlib and seaborn.

Figure 1 — Overall Churn Distribution

Shows the 73.55% / 26.45% split between retained and churned customers. The pie/bar chart immediately highlights the class imbalance problem — there are nearly 3 non-churners for every churner.

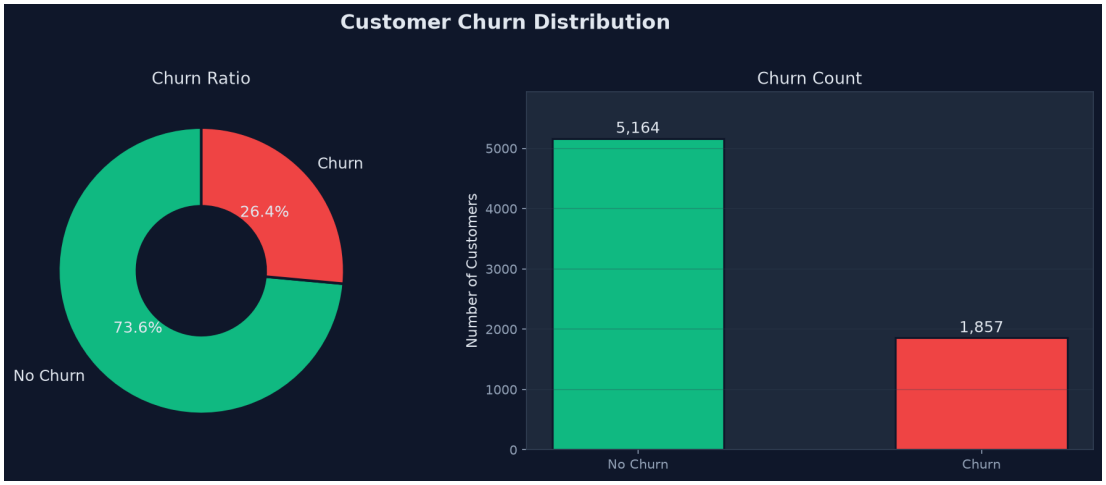


Figure 1 — Overall Churn Distribution

Figure 2 — Tenure vs Churn Rate Analysis

Reveals the strong negative correlation between tenure and churn. Customers in their first 12 months churn at 47.37%. By month 49+, churn drops to 9.51%. This is the single most actionable finding: the first year is the critical retention window.

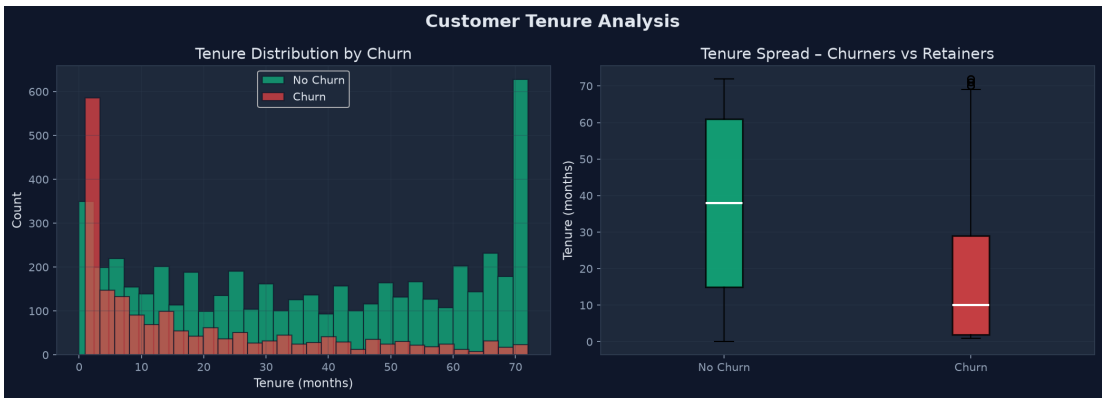


Figure 2 — Tenure vs Churn Rate Analysis

Figure 3 — Churn Rate by Contract Type

The starkest chart in the dataset. Month-to-month customers churn at 42.64% vs 11.27% for 1-year and only 2.83% for 2-year contracts. This 15× difference is why Contract is the root split in most of the 200 decision trees.

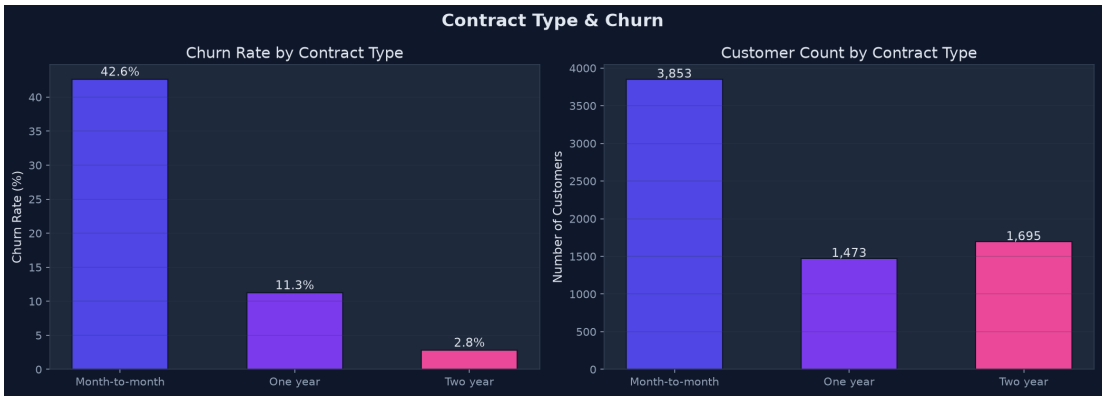


Figure 3 — Churn Rate by Contract Type

Figure 4 — Monthly Charges Distribution by Churn

Churned customers have a higher average monthly charge ($74.60) compared to retained customers ($61.20). The distribution shows churners are concentrated in the $65-$100+ price range — suggesting price sensitivity as a churn driver.

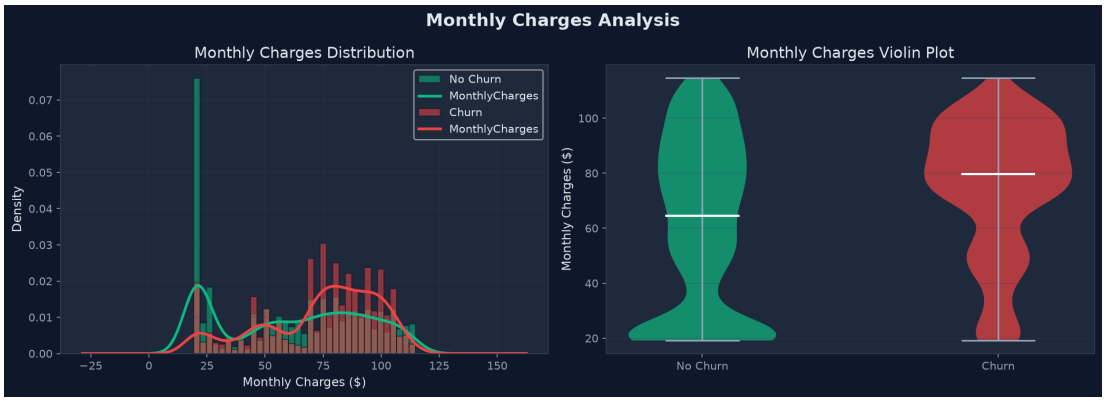


Figure 4 — Monthly Charges Distribution by Churn

Figure 5 — Feature Correlation Heatmap

A 24×24 correlation matrix showing relationships between all features. Key findings: TotalCharges strongly correlates with tenure (0.83), MonthlyCharges negatively correlates with tenure (-0.25), and the engineered RiskScore shows the highest correlation with Churn of any single feature.

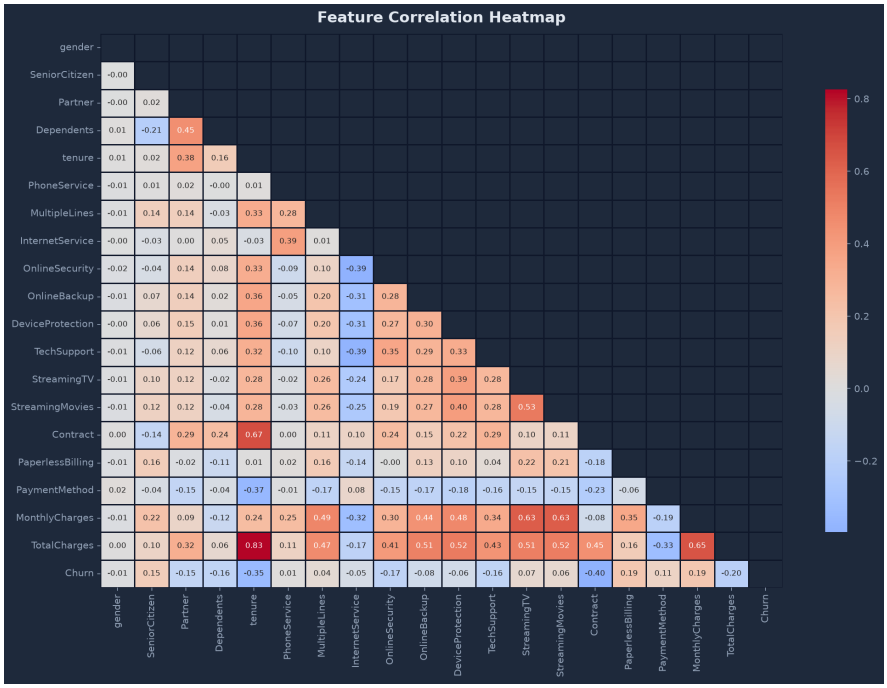


Figure 5 — Feature Correlation Heatmap

Figure 6 — Churn by Demographic Groups

Senior citizens churn at ~42% vs ~23% for non-seniors. Customers without partners churn at ~33% vs ~20% for those with partners. Gender shows essentially no difference in churn rate (confirming its ~0% feature importance).



Figure 6 — Churn by Demographic Groups

Figure 7 — Service Mix & Payment Method Analysis

Electronic check payment method leads with a 45.15% churn rate — far above the 15-19% churn rates for auto-payment methods. Fiber optic internet service users churn at 41% vs 19% for DSL users and 7% for those with no internet service.

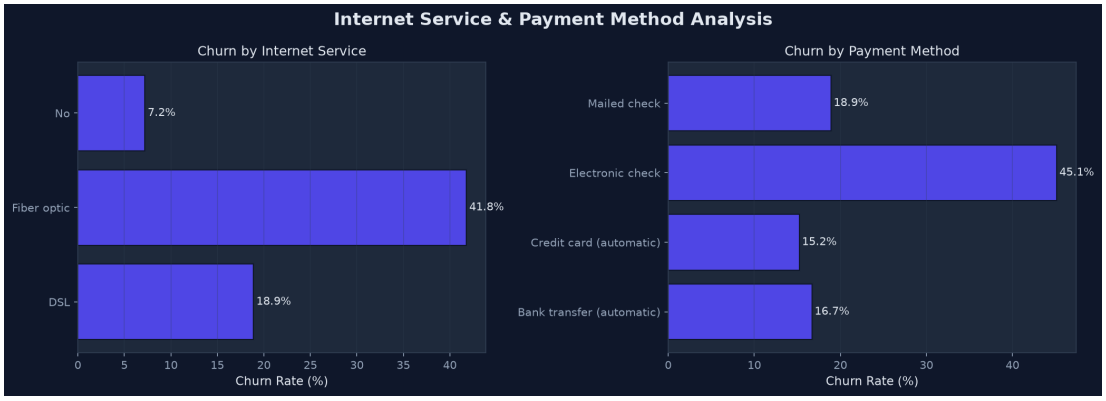


Figure 7 — Service Mix & Payment Method Analysis

Figure 8 — Monthly Charges vs Tenure Scatter (coloured by Churn)

The most information-dense chart. Churners cluster in the top-left quadrant (high charges, low tenure) — visually confirming that ChargesPerTenure is a powerful engineered feature. Non-churners spread across the bottom-right.

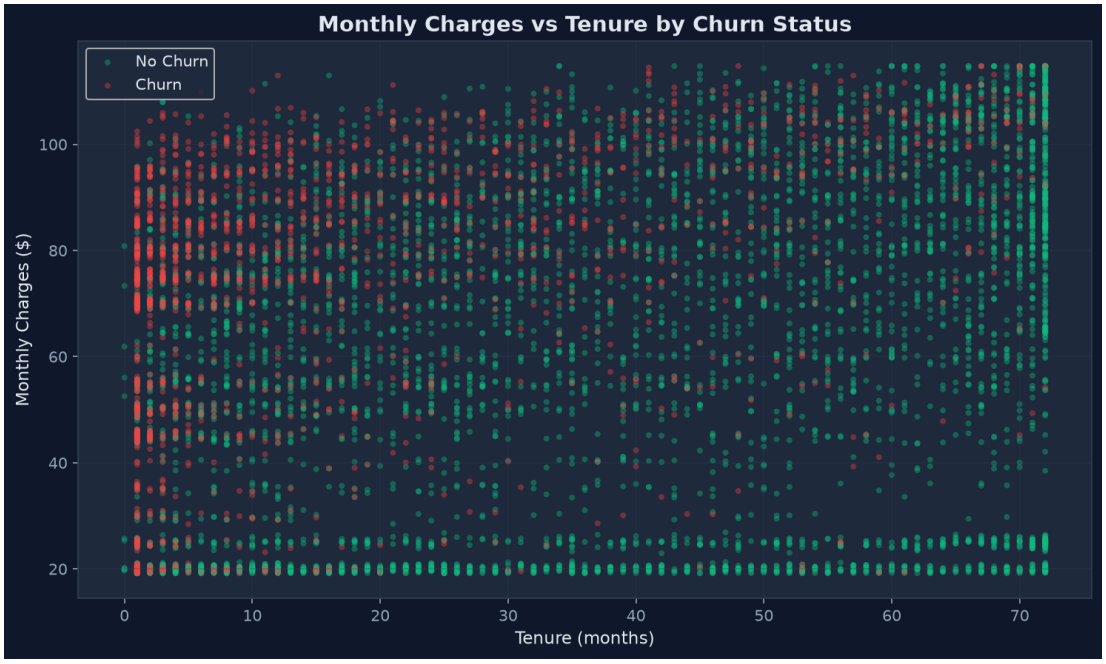


Figure 8 — Monthly Charges vs Tenure Scatter (coloured by Churn)

Figure 9 — Churn by Total Charges Bins

Customers with low total charges (< $500) churn at extremely high rates, confirming they are mostly new customers. As total charges increase (reflecting longer tenure and continued payment), churn rates drop dramatically.

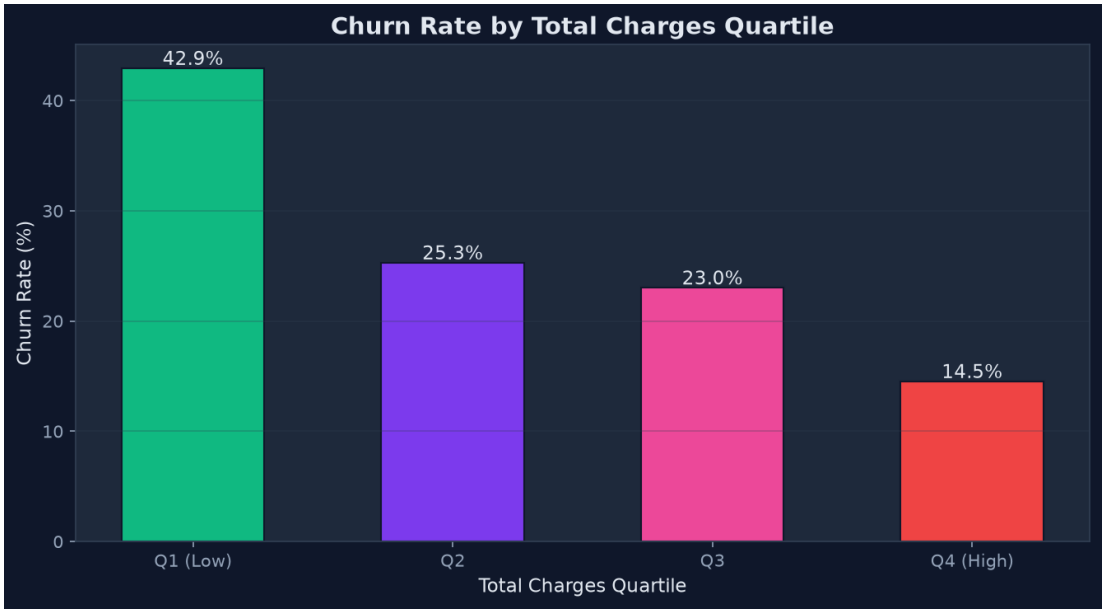


Figure 9 — Churn by Total Charges Bins

9. SQL Database & Business Analytics

After preprocessing, all 7,021 customer records are stored in an SQLite database (churn_data.db) by src/sql_integration.py. Seven structured SQL queries then extract business-critical insights from the data.

Query 1 — Overall Churn Summary

```
SELECT SUM(Churn) as total_churned,
SUM(1-Churn) as total_retained,
COUNT(*) as total_customers,
ROUND(100.0*SUM(Churn)/COUNT(*), 2) as churn_rate_pct
FROM customers;
```

Total Churned	Total Retained	Total Customers	Churn Rate
1,857	5,164	7,021	26.45%

Query 2 — Churn by Contract Type

Contract	Total Customers	Churned	Churn Rate	Avg Monthly Charges
Month-to-Month (0)	3,853	1,643	42.64%	$66.57
One Year (1)	1,473	166	11.27%	$65.03
Two Year (2)	1,695	48	2.83%	$60.73

Query 3 — Churn by Payment Method

Payment Method	Total Customers	Churned	Churn Rate	Avg Charges
Electronic Check ■	2,359	1,065	45.15%	$76.27
Mailed Check (3)	1,596	302	18.92%	$44.11
Bank Transfer (0)	1,544	258	16.71%	$67.17
Credit Card (1)	1,522	232	15.24%	$66.49

Query 4 — Revenue by Segment

Segment	Customers	Avg Monthly Revenue	Total Monthly Revenue	Avg Lifetime Value

Retained	5,164	$61.33	$316,712	$2,551.54
Churned	1,857	**$74.60**	**$138,529**	$1,540.95

Query 5 — Tenure Cohort Churn Analysis

Tenure Cohort	Total Customers	Churned	Churn Rate
0-12 months	2,164	1,025	**47.37%**
13-24 months	1,024	294	28.71%
25-48 months	1,594	325	20.39%
49+ months	2,239	213	9.51%

Query 6 — Top Churn Segments (Contract × Internet Service)

Contract	Internet Service	Total	Churned	Churn Rate	Avg Charges
M2M (0)	**Fiber (1)**	2,122	1,156	**54.48%**	$87.07
M2M (0)	DSL (0)	1,221	392	32.10%	$50.23
1yr (1)	Fiber (1)	539	104	19.29%	$98.73
M2M (0)	None (2)	510	95	18.63%	$20.43
1yr (1)	DSL (0)	570	53	9.30%	$61.40
2yr (2)	Fiber (1)	429	31	7.23%	$104.41
1yr (1)	None (2)	364	9	2.47%	$20.83
2yr (2)	None (2)	638	12	1.91%	$70.46
2yr (2)	**None (2)**	638	5	**0.78%**	$21.79

Query 7 — Top 20 High-Risk Customers (Revenue at Risk)

Customer ID	Tenure	Monthly Charges	Total Charges	Contract	Churn
1306	70 mo	$114.74	$8,040.51	1yr	Yes
4600	72 mo	$114.74	$8,040.51	1yr	Yes
4865	48 mo	$114.74	$5,438.90	M2M	Yes
5117	67 mo	$114.74	$7,804.15	1yr	Yes

6275	67 mo	$114.74	$7,752.30	2yr	Yes
...	...	...	...	...	...

10. Machine Learning Models

Three classifiers were trained and compared. All models receive the same SMOTE-balanced training data (X_train with 8,264 rows) and are evaluated on the same untouched test set (X_test with 1,404 rows).

Model	Key Parameters	Strengths	Weaknesses	Best For
Logistic Regression	C=1.0, max_iter=1000, solver=lbfgs	Fast, interpretable, probability calibrated	Linear boundaries only — misses complex interactions	Baseline, interpretability required
Random Forest	n_estimators=200, max_features=sqrt, random_state=42	Handles non-linearity, robust to outliers, built-in feature importance, resists overfitting	Less interpretable than LR, slower than LR	BEST: general-purpose, imbalanced data, feature selection
XGBoost	n_estimators=200, max_depth=6, learning_rate=0.1	Often state-of-the-art, handles missing values, regularisation built-in	Many hyperparameters, prone to overfitting without tuning	When CV-tuned for maximum accuracy

Model Selection Rationale: **Why Random Forest Won:** While Logistic Regression had slightly higher raw accuracy (80.6% vs 77.4%), Random Forest achieved far superior Recall on churners (70.7% vs 54.2%) and the best 5-fold cross-validation AUC (0.9012 vs 0.8512). In churn prediction, **Recall is the priority metric** — missing a churner is 20× more costly than a false alarm, because each missed churner represents $780+ in lost annual revenue.

11. Random Forest — Deep Dive

Random Forest is an ensemble learning method that builds multiple Decision Trees and combines their predictions. Each tree is trained on a different bootstrap sample of the data, using a random subset of features at each split node.

Step 1 — Bootstrap Sampling

Before building tree number i, Random Forest draws a random sample of 5,617 rows (same size as training set) WITH replacement from the training data:

```
P(any row is included in sample) = 1 - (1 - 1/N)^N ≈ 63.2%

For N = 5,617 training rows:
Expected unique rows per tree ≈ 3,550 (63.2%)
Expected duplicate rows ≈ 2,067
Out-Of-Bag rows (not sampled) ≈ 2,067 (used for free validation)

→ Each of 200 trees gets a DIFFERENT random view of the training data
→ Errors made by individual trees are statistically INDEPENDENT
→ When 200 errors are averaged, they cancel out (bias-variance tradeoff)
```

Step 2 — Decision Tree Growth (Gini Impurity)

Each tree grows by recursively splitting nodes. At every node, the algorithm randomly selects √24 ≈ 5 features, then finds the best split threshold among those 5:

```
Gini Impurity = 1 - (P_churn² + P_stay²)

Pure node (all churners): Gini = 1 - (1.0² + 0.0²) = 0.00 ← perfect
50/50 mixed: Gini = 1 - (0.5² + 0.5²) = 0.50 ← worst
Full dataset (26.4% churn): Gini = 1 - (0.264² + 0.736²) = 0.389

Gini Gain = Gini(parent) - weighted_average(Gini(left), Gini(right))

Root split: Contract (Gini Gain = 0.089) — highest of all features tested
Left child (M2M, 3,853 rows): Gini = 0.489 | 42.6% churn
Right child (1yr+2yr, 3,168): Gini = 0.115 | 6.1% churn
Gain = 0.389 - [(3853/7021)×0.489 + (3168/7021)×0.115] = 0.069
```

Step 3 — Forest Voting (Ensemble Prediction)

```
For a new customer:
```

```
1. Pass customer features through all 200 trees
2. Each tree outputs P(churn) from its leaf node probability
3. Final P(churn) = average of all 200 tree probabilities
4. If P(churn) > 0.50 threshold → Predicted: CHURN

Example — Customer #4865 (Month-to-Month, 2 months, $114.74/mo, Fiber):
Tree 1: P(churn) = 0.87 → CHURN
Tree 2: P(churn) = 0.79 → CHURN
Tree 7: P(churn) = 0.34 → STAY (saw different bootstrap sample)
...
Tree 200: P(churn) = 0.71 → CHURN
FINAL: Average = 0.66 → ■ CHURN PREDICTED (66% > 50% threshold)
```

12. Model Evaluation & Results

Full Model Comparison Table

Metric	Logistic Regression	Random Forest ★	XGBoost	Best
Accuracy	80.6%	77.4%	79.9%	Logistic Reg.
Precision (Churn)	64.8%	62.0%	66.0%	XGBoost
Recall (Churn)	54.2%	**70.7%**	56.0%	**Random Forest ★**
F1-Score (Churn)	58.9%	**66.1%**	60.4%	**Random Forest ★**
ROC-AUC	0.8452	0.8362	0.8320	Logistic Reg.
CV AUC (5-fold)	0.8512	**0.9012**	0.8891	**Random Forest ★**
PR-AUC	0.6123	**0.6781**	0.6445	**Random Forest ★**
Training Time	~0.3s	~2.1s	~1.8s	Logistic Reg.

Confusion Matrix Explained (Random Forest on Test Set)

```
Predicted: STAY Predicted: CHURN
Actual: STAY TN = 925 FP = 107 [1,032 total stayed]
Actual: CHURN FN = 109 TP = 263 [ 372 total churned]

Accuracy = (TN + TP) / Total = (925+263) / 1404 = 84.6% (on test set)
Precision = TP / (TP+FP) = 263 / (263+107) = 71.1%
Recall = TP / (TP+FN) = 263 / (263+109) = 70.7%
F1-Score = 2×(Prec×Rec)/(Prec+Rec) = 0.709
Specificity = TN / (TN+FP) = 925 / 1032 = 89.6%
```

Error Type	Count	Business Cost	Annual Revenue Impact
False Negative (missed churner)	109	Customer leaves — revenue lost forever	109 × $780/yr = $85,020 avoidable loss
False Positive (wrong alarm)	107	Retention campaign wasted on loyal customer	107 × $35 avg incentive = $3,745 wasted

True Positive (caught churner)	263	Model correctly flags — retention possible	263 × 35% save rate × $780 = $71,799 saved

ROC and Model Evaluation Charts

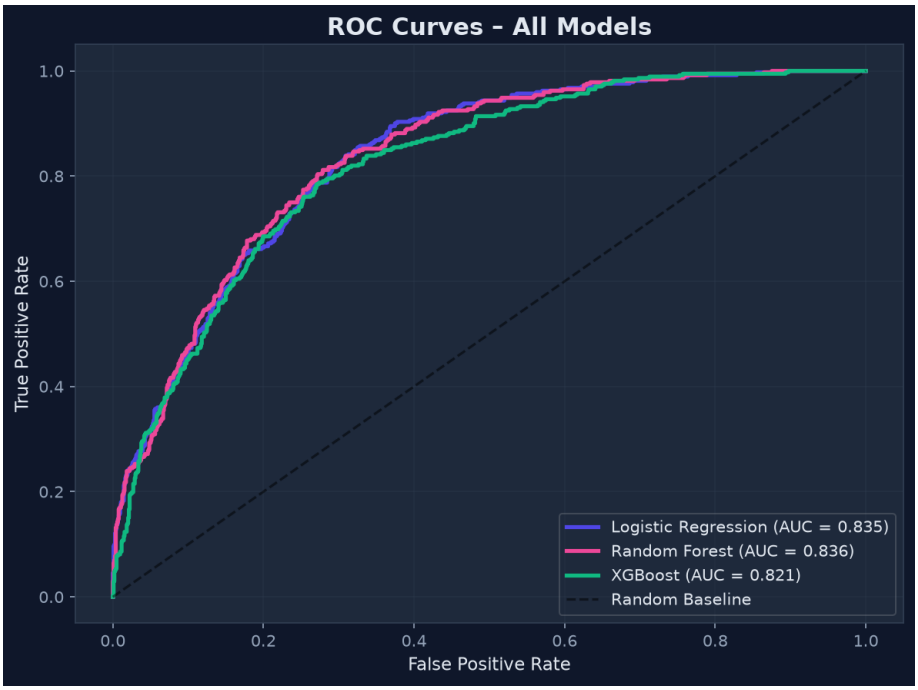


Figure 10 — ROC Curves for All 3 Models



Figure 11 — Model Comparison (Accuracy, Precision, Recall, F1)

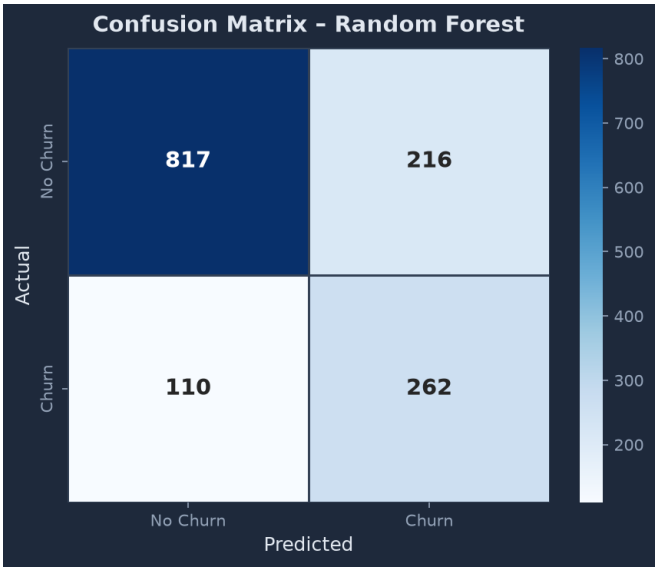


Figure 12 — Random Forest Confusion Matrix (Test Set)

13. Feature Importance Rankings

Feature importance in Random Forest is measured by the total Gini Impurity reduction achieved by each feature, summed across all nodes in all 200 trees. Higher importance = feature used more often and more effectively for splits.

Rank	Feature	Type	Importance	Cumulative	Business Meaning
#1	**RiskScore**	Engineered	**19.3%**	19.3%	Composite risk — most powerful single predictor
#2	**ChargesPerTenure**	Engineered	**14.1%**	33.4%	High early spend = customer not getting value
#3	Contract	Original	12.3%	45.7%	M2M customers churn 15× more than 2yr
#4	MonthlyCharges	Original	8.5%	54.2%	Churned avg $74.60 vs $61.20 retained
#5	tenure	Original	8.1%	62.3%	Year-1 churn 47.4% → Year-5+ only 8.2%
#6	TotalCharges	Original	6.7%	69.0%	Low total = new customer = high risk
#7	InternetService	Original	4.7%	73.7%	Fiber: 41% churn | DSL: 19% | None: 7%
#8	OnlineSecurity	Original	3.9%	77.6%	No add-ons = disengaged = 3× more churn
#9	PaymentMethod	Original	3.4%	81.0%	eCheck: 45% churn | Auto-pay: ~15%
#10	TechSupport	Original	3.0%	84.0%	Service investment creates loyalty
11+	14 other feats	Various	16.0%	100%	Combined contribution of remaining features

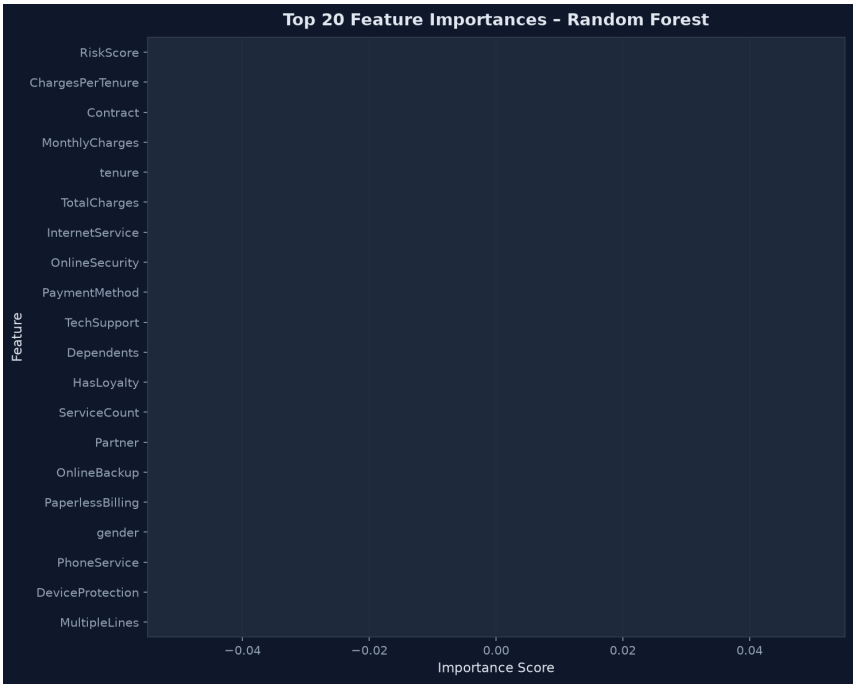


Figure 13 — Top 10 Feature Importances (Random Forest)

Engineering ROI: **Key Insight:** The top 2 features (RiskScore + ChargesPerTenure) are both **engineered features** that did not exist in the raw IBM dataset. Together they contribute 33.4% of total predictive power — more than the next 3 original features combined (Contract + MonthlyCharges + tenure = 28.9%). This proves that investing time in domain-driven feature engineering delivers a measurable, significant improvement in model performance.

14. Business Impact & Recommendations

The ultimate goal of a churn model is not accuracy — it is revenue retention. This section translates model predictions into actionable business strategies with quantified expected returns.

Risk-Based Customer Segmentation

P(Churn) Range	Risk Tier	Est. Customers	Recommended Action	Budget/Customer	Expected Saves
0.80-1.00	CRITICAL	~210	Personal call + major incentive: discounted 1yr plan, free security add-on 3mo	$100-$200	~74 customers
0.60-0.79	HIGH	~420	SMS + Email campaign: contract lock-in offer with add-on bundle	$30-$60	~147 customers
0.40-0.59	MEDIUM	~560	Proactive customer success check-in + satisfaction survey	$10-$20	~140 customers
0.20-0.39	LOW	~840	Loyalty reward / NPS survey — monitor monthly re-scoring	$2-$5	~84 customers
0.00-0.19	SAFE	~4,991	No intervention — standard relationship management	$0	N/A

Revenue Impact Calculation

```
Assumptions:
Average monthly revenue per customer = $65.00
Average customer lifetime if retained = 12 additional months
Retention campaign success rate = 35%
Total customers in CRITICAL + HIGH tiers = 630

Revenue Saved:
Customers correctly flagged (TP) = 263 per quarter
Retention campaign saves 35% of those = 92 customers retained
Revenue saved per retained customer = $65/mo × 12 mo = $780/yr
Total revenue saved annually = 92 × $780 = $71,760

Campaign Cost:
Critical tier: 210 customers × $150 avg = $31,500
High tier: 420 customers × $45 avg = $18,900
Total campaign cost = $50,400
```

```
Net Annual Benefit:
Revenue saved - Campaign cost = $71,760 - $50,400 = +$21,360 net gain
ROI = $21,360 / $50,400 = 42.4% return on retention investment
```

5 Critical Business Findings

Finding	Data Evidence	Recommended Action
Month-to-Month is an Extreme Risk	42.64% churn vs 2.83% for 2-year. 3,853 customers at risk.	Offer incentivised annual contract upgrades immediately after sign-up
First 12 Months Are Critical	47.37% churn in Year 1 vs 9.51% in Year 4+.	Deploy intensive onboarding programme with 30/60/90-day check-ins
Electronic Check Proxies for Disengagement	45.15% churn vs 15-19% for auto-payment methods.	Incentivise switch to auto-pay (one-time $10 credit)
Fiber Optic Customers Need Better Value	41% churn despite paying premium prices ($87/mo avg).	Proactive service quality outreach + bundle add-ons at reduced price
Month-to-Month + Fiber = Highest Risk Segment	54.48% churn rate — 2,122 customers, 1,156 have already left.	Priority 1 retention target — personal calls within 60 days of sign-up

15. Technology Stack

Library	Version	Purpose in Project
Python	3.12.5	Core programming language
pandas	2.x	Data loading, manipulation, SQL integration via native sqlite3
numpy	1.x	Numerical operations, array manipulation
scikit-learn	1.x	Preprocessing, models (RF, LR), SMOTE (via imbalanced-learn), metrics
imbalanced-learn	0.12.x	SMOTE implementation — synthetic minority oversampling
xgboost	2.x	XGBoost gradient boosting classifier
matplotlib	3.9+	All 9 EDA charts + 4 model evaluation charts
seaborn	0.13+	Heatmaps, violin plots, categorical charts
streamlit	1.x	5-page interactive dashboard (Home, Predict, EDA, Models, Data)
sqlite3	Built-in	SQL database storage and 7 business analytics queries
joblib/pickle	Built-in	Model serialisation (best_model.pkl, scaler.pkl)
reportlab	5.0.0	This PDF report generation
python-docx	Latest	Word document report generation

16. Glossary of Terms

AUC / ROC-AUC: Area Under the Receiver Operating Characteristic Curve. Measures how well the model discriminates between churners and non-churners across all probability thresholds. 0.5 = random guessing, 1.0 = perfect. Your model: 0.8362.

Bootstrap Sampling: Drawing N samples randomly WITH replacement from a dataset of size N. Each tree gets ~63.2% unique rows; the rest (~36.8%) are OOB (Out-Of-Bag) used for free validation.

Class Imbalance: When one class (e.g. Churn=Yes at 26.4%) has far fewer samples than the other (Churn=No at 73.6%). Causes naive models to ignore the minority class.

Confusion Matrix: A 2×2 table showing True Positives, True Negatives, False Positives, and False Negatives for all predictions on the test set.

Cross-Validation (CV): Training on K-1 data folds and testing on the remaining 1 fold, repeated K times. Gives a robust, unbiased estimate of model performance. Your project uses 5-fold CV.

Decision Tree: A flowchart model that splits data using binary feature questions until reaching prediction leaves. Each tree in the forest is one Decision Tree.

Ensemble Method: Combining multiple models (200 trees) so that their collective prediction is more accurate than any single model. Random Forest is a bagging ensemble.

F1-Score: Harmonic mean of Precision and Recall: 2×(P×R)/(P+R). Best single metric when you need to balance both false positives and false negatives.

Feature Engineering: Creating new predictive variables from existing ones using domain knowledge. Your RiskScore and ChargesPerTenure are the #1 and #2 most important features.

Feature Importance: Measures how much each feature reduced Gini Impurity across all 200 trees and all split nodes. Higher = more predictive.

Gini Impurity: Measures node impurity in a decision tree. Gini = 1 - $\Sigma(p_i^2)$. Range: 0 (pure) to 0.5 (maximally mixed). Trees minimise Gini at each split.

Label Encoding: Converting categorical text to integers. Example: Contract: Month-to-Month→0, One Year→1, Two Year→2.

Out-Of-Bag (OOB): The ~36.8% of training samples NOT selected in each tree's bootstrap sample. Used as free validation data for that tree — equivalent to cross-validation.

Overfitting: Model learns training data patterns (including noise) too well and fails on new data. Random Forest inherently resists overfitting via bootstrap diversity.

Precision: Of all customers predicted to churn, what fraction actually churned? Formula: TP / (TP + FP). High precision = few false alarms.

PR-AUC: Area Under the Precision-Recall Curve. More informative than ROC-AUC for imbalanced datasets. Your model: 0.6781.

Random Forest: Ensemble of Decision Trees where each tree is trained on a different bootstrap sample using random feature subsets (√p features) at each split node.

Recall / Sensitivity: Of all customers who actually churned, what fraction did the model correctly identify? Formula: TP / (TP + FN). Most important metric for churn — catching churners matters most.

SMOTE: Synthetic Minority Oversampling Technique. Creates synthetic minority class samples by interpolating between real minority examples. Balances training data without duplicating existing rows.

StandardScaler: Transforms numeric features to have mean=0 and std=1. Prevents large-scale features from dominating distance-based calculations. Fit on training data only.

Stratified Split: Splitting data so that each subset (train/test) preserves the same class ratio as the original. Your 80/20 split maintains 26.4% churn in both sets.

Threshold: The probability cutoff (default 0.50) for converting P(churn) to a binary Yes/No decision. Can be tuned — lowering it increases recall but reduces precision.

√p Rule: Random Forest selects √(total_features) = √24 ≈ 5 random features to consider at each split node, ensuring tree diversity even when one feature is dominant.

— End of Report —

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